

Surya S

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Professional Summary

M.Tech. Computer Science and Engineering student at **NIT Calicut** with 10 months of production experience at **SAP Labs**, building scalable backend services and AI-powered applications. Skilled in **Java, Python, FastAPI, REST APIs, React.js, Node.js, MongoDB, RAG, LLMs, LangChain, and LangGraph**. Delivered a **98% reduction in execution time** for a critical production workflow.

Education

National Institute of Technology Calicut

M.Tech. in Computer Science and Engineering – Information Security | CGPA: 7.58/10

Kozhikode, Kerala, India

2024 – 2026

Government College of Engineering

B.E. in Computer Science and Engineering | CGPA: 8.66/10

Bargur, Tamil Nadu, India

2020 – 2024

Experience

SAP Labs India

Software Developer Intern

Bengaluru, India

August 2025 – June 2026

- Engineered scalable backend services for an enterprise **RAG platform** supporting document grounding and knowledge retrieval using **Python, FastAPI, REST APIs, and microservices**.
- Designed and optimized REST APIs for document ingestion and retrieval workflows, enabling integration with **LLM-powered services** and improving backend processing efficiency.
- Redesigned a critical production data-deletion workflow, reducing execution time from **300+ seconds to under 4 seconds** and achieving a **98% reduction in execution time**.
- Collaborated with cross-functional Agile teams to troubleshoot production issues and improve service reliability; contributed to **Docker-based development, CI/CD workflows, and cloud-native microservices**.

Technical Skills

Languages: Java, Python, C

Backend: FastAPI, Node.js, Express.js, REST APIs, Microservices

Frontend: React.js, JavaScript, HTML5, CSS3, Tailwind CSS

Databases: MongoDB, MySQL, SQLite, SAP HANA

AI/GenAI: LLMs, RAG, LangChain, LangGraph, Agentic AI, Generative AI, Prompt Engineering

DevOps/Tools: Git, Docker, Kubernetes, CI/CD, FAISS, ChromaDB, Streamlit, Gradio

Core CS: Data Structures and Algorithms, Object-Oriented Programming, System Design Fundamentals

Projects

Agentic AI Chatbot with Retrieval-Augmented Generation (RAG) *Python, LangGraph, LangChain, FAISS, SQLite* Live Demo

- Built an autonomous **Agentic AI chatbot** using LangGraph with persistent checkpoint-based memory, dynamic tool calling, and multi-session conversation management.
- Designed a **RAG pipeline** using vector embeddings, FAISS, and semantic search to retrieve relevant context from custom knowledge bases for grounded LLM responses.
- Integrated external APIs and developed a Streamlit interface supporting **streaming responses, chat history, and session-based conversations**.

Resume Building Platform with AI Assistance *React.js, Tailwind CSS, Express.js, MongoDB, JWT, Gemini API*

GitHub

- Developed a full-stack resume-building platform with predefined templates and **AI-assisted content enhancement** using the Google Gemini API.
- Implemented **JWT-based authentication and authorization**, designed RESTful APIs with Express.js, and integrated MongoDB for dynamic resume generation and data management.

TextBook GPT *Python, LangChain, ChromaDB, Self-RAG, Agentic RAG*

- Built a **Self-RAG based textbook question-answering system** that evaluates retrieved context for relevance and adaptively decides whether additional retrieval is required before generating responses.
- Integrated **web search with the retrieval workflow** to dynamically supplement knowledge-base content when retrieved information is insufficient, improving contextual accuracy and response quality.

Achievements & Certifications

GATE 2024: Ranked among the **top 4% nationally** in GATE (Computer Science and Information Technology).

Certifications: Retrieval Augmented Generation – Krish Naik (Udemy) | MERN Full Stack Development – E-Learn Infotech